High levels of carcinogenic polycyclic aromatic hydrocarbons in mate drinks.

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BACKGROUND: Drinking mate has been associated with cancers of the esophagus, oropharynx, larynx, lung, kidney, and bladder. We conducted this study to determine whether drinking mate could lead to substantial exposure to polycyclic aromatic hydrocarbons (PAH), including known carcinogens, such as benzo[a]pyrene. METHODS: The concentrations of 21 individual PAHs were measured in dry leaves of eight commercial brands of yerba mate and in infusions made with hot (80 degrees C) or cold (5 degrees C) water. Measurements were done using gas chromatography/mass spectrometry, with deuterated PAHs as the surrogates. Infusions were made by adding water to the leaves, removing the resulting infusion after 5 min, and then adding more water to the remaining leaves. This process was repeated 12 times for each infusion temperature. RESULTS: The total concentrations of the 21 PAHs in different brands of yerba mate ranged from 536 to 2,906 ng/g dry leaves. Benzo[a]pyrene concentrations ranged from 8.03 to 53.3 ng/g dry leaves. For the mate infusions prepared using hot water and brand 1, 37% (1,092 of 2,906 ng) of the total measured PAHs and 50% (25.1 of 50 ng) of the benzo[a]pyrene content were released into the 12 infusions. Similar results were obtained for other hot and cold infusions. CONCLUSION: Very high concentrations of carcinogenic PAHs were found in yerba mate leaves and in hot and cold mate infusions. Our results support the hypothesis that the carcinogenicity of mate may be related to its PAH content.